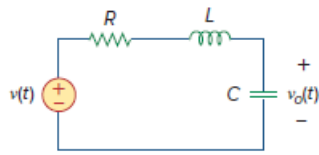


Problem

Using Fig. 17.70, design a problem to help other students better understand circuit responses to a Fourier series.

Figure 17.70:



Step-by-step solution

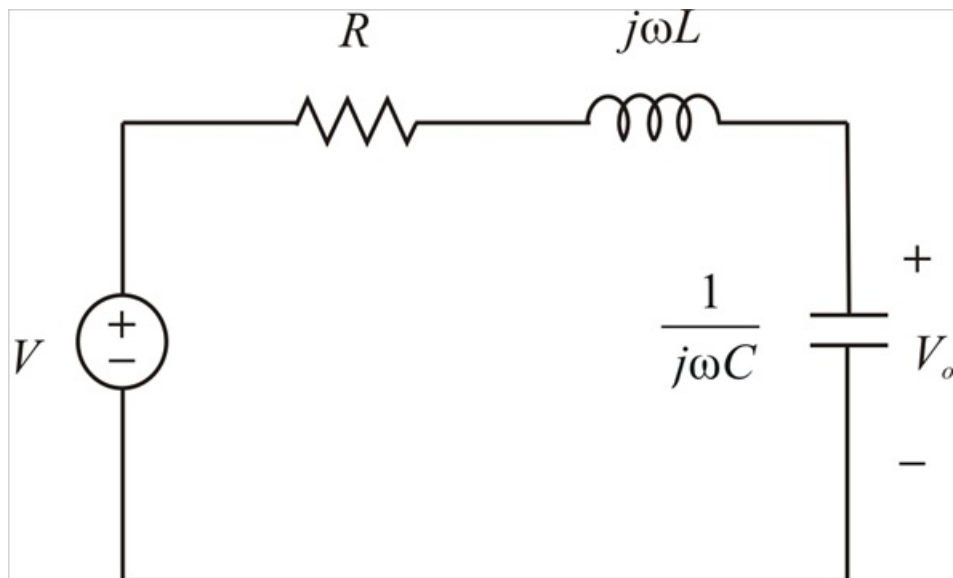
Step 1 of 4

Consider the input signal as

$$v(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) \text{ V}$$

Step 2 of 4

The circuit in frequency domain is,



Step 3 of 4

Where V is the phasor form of supply voltage $v(t)$, by using the voltage division rule,

$$V_o = \frac{\frac{1}{j\omega_n C}}{\frac{1}{j\omega_n C} + R + j\omega_n L} V$$

$$V_o = \frac{1}{1 + j\omega_n RC - (\omega_n LC)^2} V$$

For dc or $\omega_n = 0$,

$$v_o(0) = v(0)$$

We know $v(0) = a_0$

$$\therefore v_o(0) = a_0$$

Step 4 of 4

For the n th harmonic,

$$V = \sqrt{a_n^2 + b_n^2} \angle \tan^{-1} \left(\frac{b_n}{a_n} \right)$$

And the corresponding response is,

$$V_o = \frac{1}{\sqrt{\left(1 - (n\omega_0 LC)^2\right)^2 + (n\omega_0 RC)^2}} \angle \tan^{-1} \left(\frac{n\omega_0 RC}{1 - n^2 \omega_0^2 L^2 C^2} \right) \left(\sqrt{a_n^2 + b_n^2} \angle \tan^{-1} \left(\frac{b_n}{a_n} \right) \right)$$

$$V_o = \frac{\sqrt{a_n^2 + b_n^2}}{\sqrt{\left(1 - (n\omega_0 LC)^2\right)^2 + (n\omega_0 RC)^2}} \angle \left[\tan^{-1} \left(\frac{b_n}{a_n} \right) - \tan^{-1} \left(\frac{n\omega_0 RC}{1 - n^2 \omega_0^2 L^2 C^2} \right) \right]$$

Therefore, the output voltage in time domain is,

$$v_o(t) = a_0 + \frac{\sqrt{a_n^2 + b_n^2}}{\sqrt{\left(1 - (n\omega_0 LC)^2\right)^2 + (n\omega_0 RC)^2}} \angle \left[\tan^{-1} \left(\frac{b_n}{a_n} \right) - \tan^{-1} \left(\frac{n\omega_0 RC}{1 - n^2 \omega_0^2 L^2 C^2} \right) \right]$$